GENERATIVE AI INFRASTRUCTURE · PORTFOLIO ARTIFACT

What it takes to train a large language model

From raw text to a deployed chatbot: the six steps of LLM training, and the data, compute, energy, time and people each one uses.

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Why training costs matter

Training one frontier model can cost **tens to hundreds of millions of dollars**, and the bill grows about **2.4× every year**.

PART 1

The training pipeline

Six steps, from collecting data to serving users

PART 2

The four resources

Data, compute, energy, and time & people

PART 3

Real models, real numbers

GPT-4, Claude, LLaMA, Gemini and DeepSeek

Source: Cottier et al. (2024), Epoch AI

The training pipeline

Every major LLM follows roughly the same six steps. Pre-training uses most of the money, energy and time.

1. Collect data

Trillions of words: web pages, books, code, licensed content

Data · People

2. Clean & tokenize

Remove spam, duplicates and personal data; split text into tokens

Data · Compute

3. Pre-train

Predict the next token across the whole dataset on thousands of GPUs for weeks to months

Biggest cost

4. Fine-tune

Train on curated question-and-answer examples written by people

People · Compute

5. Align & test

Humans rank answers (RLHF); red teams probe for harmful output

People · Compute

6. Evaluate & deploy

Benchmark, then serve users; running it (inference) costs money daily

Compute · Energy

Prepare the data (1–2) → Learn language (3) → Make it helpful, safe and available (4–6)

The four resources behind every model

Data

~15T

tokens used to train Llama 3.1, about 50× GPT-3

Compute

30.8M

H100 GPU-hours for Llama 3.1 405B

Energy

1,287

MWh to train GPT-3, about 120 U.S. homes for a year

Time & people

~80

days of nonstop work by 16,000 GPUs*

Each resource drives cost differently, and each one shapes how capable the finished model is.

Sources: Meta (2024); Patterson et al. (2021); Brown et al. (2020) · *Author's calculation

Data: the text a model learns from

~15T

tokens to train Llama 3.1 (2024), about 50× GPT-3's 300 billion (2020)

Free to collect, costly to clean

Most web data costs nothing to scrape, but filtering spam, duplicates and personal data takes large engineering teams.

Quality data now has a price

Google reportedly pays Reddit about **$60M a year** for access to its content.

Impact on performance: more, cleaner and more diverse data gives broader knowledge and fewer errors.

Sources: Meta (2024); Brown et al. (2020); Tong, Wang & Coulter (2024), Reuters

Compute: the biggest cost driver

30.8M

H100 GPU-hours to train Llama 3.1 405B, on up to 16,000 GPUs

Tens of millions per run

One H100 GPU-hour rents for roughly $2 or more, so a large training run costs tens of millions of dollars.

Chips are the largest item

AI accelerator chips are about **44%** of a frontier model's final training-run cost.

Impact on performance: more compute allows bigger models trained on more data, the main driver of capability gains.

Sources: Meta (2024); Grattafiori et al. (2024); Cottier et al. (2024), Epoch AI

Energy: small share of cost, big footprint

1,287

MWh to train GPT-3, about what 120 U.S. homes use in a year (552 t CO_2e)

Emissions grow with scale

Training the Llama 3.1 family produced an estimated **11,390 t CO_2e** (location-based); Meta matched it with renewable power.

Only 2–6% of the bill

Energy is a small part of total development cost, but its environmental impact is large.

Impact on performance: energy scales with compute, so efficient hardware and code get more training per dollar.

Sources: Patterson et al. (2021); Meta (2024); Cottier et al. (2024), Epoch AI

Time & people: the hidden costs

~80 days

of nonstop work by 16,000 GPUs to pre-train Llama 3.1 405B*

People are a major cost

Researchers, engineers and human raters make up **29–49%** of total development cost.

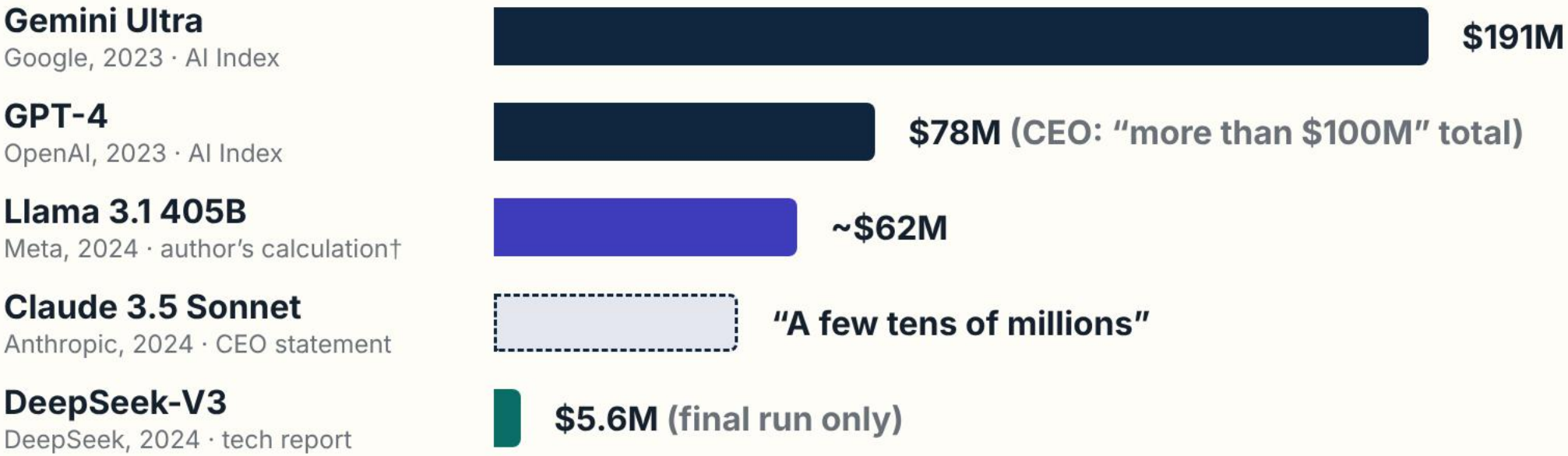
Months before the final run

Failed experiments and smaller test runs come before the final run even begins.

Impact on performance: skilled teams and human feedback turn a raw model into a helpful, safe assistant.

*Author's calculation: 30.84M GPU-hours ÷ 16,000 GPUs · Source: Cottier et al. (2024), Epoch AI

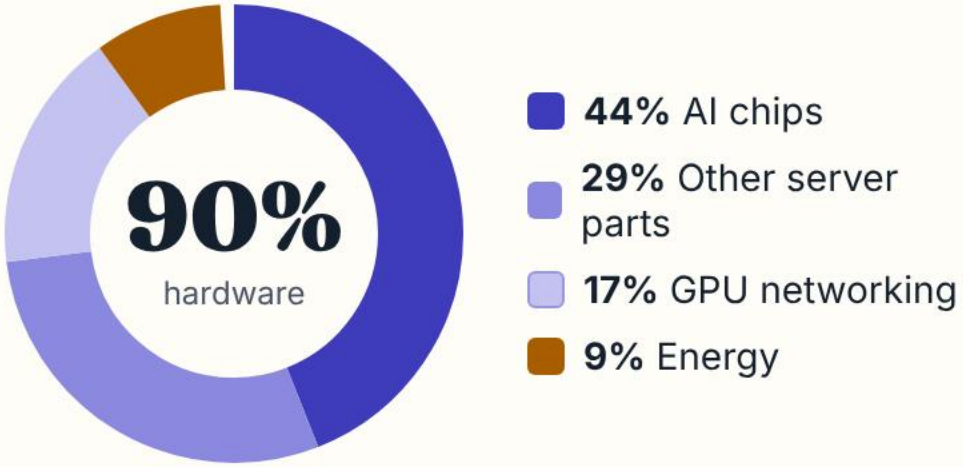
Estimated cost of the final training run



Bar length = cost; dashed = range (not disclosed). †30.84M GPU-hours × $2/hour. Methods differ; not directly comparable.

Where the money goes, and how to spend less

Final-run cost breakdown



Similar data, 11× less compute

GPU-hours for the full training run, both on about 15 trillion tokens



DeepSeek's mixture-of-experts design activates only part of the model for each token.

Sources: Cottier et al. (2024), Epoch AI; Meta (2024); DeepSeek-AI (2024)

What each model used

Model	Training data	Compute
GPT-3 (OpenAI, 2020)	300B tokens	175B parameters
Llama 3.1 405B (Meta, 2024)	~15T tokens	30.8M H100 GPU-hours
DeepSeek-V3 (2024)	14.8T tokens	2.8M H800 GPU-hours, 2,048 GPUs
GPT-4, Gemini, Claude	Not disclosed	Not disclosed

The leading proprietary labs publish almost none of these numbers, so their costs come from independent estimates and executive statements.

Sources: Brown et al. (2020); Meta (2024); DeepSeek-AI (2024)

Key takeaways

1

Costs are rising fast

Frontier training costs have grown about 2.4× per year since 2016.

2

Headlines hide most of the bill

Reported costs usually cover only the final run, not experiments, staff or data deals.

3

Efficiency matters

DeepSeek-V3 shows smart engineering can sharply cut the cost of a strong model.

Sources: Cottier et al. (2024), Epoch AI; DeepSeek-AI (2024)

What the costs mean for you

Building AI products

Training from scratch is out of reach for most companies; fine-tuning an existing model costs a tiny fraction.

Investing

Compute, chips and energy contracts are the main barriers to entry, so a few well-funded labs lead.

Setting policy or ESG goals

Energy and emissions grow with each model generation; ask vendors how they measure and offset them.

Working in AI or cloud

Most projects adapt existing models, so fine-tuning, retrieval, cloud and MLOps skills are in high demand.

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Author's calculations: ~80 days = 30.84M GPU-hours ÷ 16,000 GPUs; ~$62M = 30.84M GPU-hours × $2/hour. Cost estimates use different methods and are not directly comparable.